

Top 100 Diagnosis Features by Intersection

[4 methods] Elastic Net n LightGBM n NTK n Recurrence Enrichment (n=15)

B18: Chronic viral hepatitis
B19: Unspecified viral hepatitis
D57: Sickle-cell disorders
F10: Alcohol related disorders
F11: Opioid related disorders
F14: Cocaine related disorders
F17: Nicotine dependence
F19: Other psychoactive substance related disorders
F31: Bipolar disorder
F32: Major depressive disorder, single episode
F55: Abuse of non-dependence-producing substances
G89: Pain, not elsewhere classified
O99: Other maternal diseases classifiable elsewhere but complicat
R45: Symptoms and signs involving emotional state
T40: Poisoning by, adverse effect of and underdosing of narcotics

[3 methods] LightGBM n NTK n Recurrence Enrichment (n=20)

E66: Overweight and obesity
F41: Other anxiety disorders
J44: Other chronic obstructive pulmonary disease
J45: Asthma
K21: Gastro-esophageal reflux disease
L03: Cellulitis and acute lymphangitis
M54: Dorsalgia
M79: Other and unspecified soft tissue disorders, not elsewhere c
R00: Abnormalities of heart beat
R06: Abnormalities of breathing
R07: Pain in throat and chest
R10: Abdominal and pelvic pain
R11: Nausea and vomiting
R41: Other symptoms and signs involving cognitive functions and a
Y92: Place of occurrence of the external cause
Z68: Body mass index [BMI]
Z79: Long term (current) drug therapy
Z86: Personal history of certain other diseases
Z88: Allergy status to drugs, medicaments and biological substanc
Z91: Personal risk factors, not elsewhere classified

[3 methods] Elastic Net n LightGBM n NTK (n=9)

E78: Disorders of lipoprotein metabolism and other lipidemias
I48: Atrial fibrillation and flutter
K57: Diverticular disease of intestine
K80: Cholelithiasis
N20: Calculus of kidney and ureter
Z01: Encounter for other special examination without complaint, s
Z12: Encounter for screening for malignant neoplasms
Z37: Outcome of delivery
Z3A: Weeks of gestation

[3 methods] Elastic Net n LightGBM n Recurrence Enrichment (n=3)

B20: Human immunodeficiency virus [HIV] disease
K86: Other diseases of pancreas
Z76: Persons encountering health services in other circumstances

[2 methods] Elastic Net n LightGBM (n=30)

A02: Other salmonella infections
C50: Malignant neoplasm of breast
D25: Leiomyoma of uterus
F02: Dementia in other diseases classified elsewhere
F03: Unspecified dementia
H25: Age-related cataract
H26: Other cataract
H65: Nonsuppurative otitis media
I21: ST elevation (STEMI) and non-ST elevation (NSTEMI) myocardia
I65: Occlusion and stenosis of precerebral arteries, not resultin
I70: Atherosclerosis
J35: Chronic diseases of tonsils and adenoids
K08: Other disorders of teeth and supporting structures
K35: Acute appendicitis
K40: Inguinal hernia
K64: Hemorrhoids and perianal venous thrombosis
N47: Disorders of prepuce
N83: Noninflammatory disorders of ovary, fallopian tube and broad
N84: Polyp of female genital tract
N92: Excessive, frequent and irregular menstruation
O09: Supervision of high risk pregnancy
O70: Perineal laceration during delivery
S42: Fracture of shoulder and upper arm
S52: Fracture of forearm
S62: Fracture at wrist and hand level
S83: Dislocation and sprain of joints and ligaments of knee
Z02: Encounter for administrative examination
Z30: Encounter for contraceptive management
Z53: Persons encountering health services for specific procedures
Z66: Do not resuscitate

[2 methods] NTK n Recurrence Enrichment (n=25)

A41: Other sepsis
D50: Iron deficiency anemia
D63: Anemia in chronic diseases classified elsewhere
D64: Other anemias
D69: Purpura and other hemorrhagic conditions
D72: Other disorders of white blood cells
E83: Disorders of mineral metabolism
E86: Volume depletion
E87: Other disorders of fluid, electrolyte and acid-base balance
F12: Cannabis related disorders
F43: Reaction to severe stress, and adjustment disorders
G40: Epilepsy and recurrent seizures
G43: Migraine
G47: Sleep disorders
G93: Other disorders of brain
I95: Hypotension
J18: Pneumonia, unspecified organism
J96: Respiratory failure, not elsewhere classified
K59: Other functional intestinal disorders
M47: Spondylosis
M51: Thoracic, thoracolumbar, and lumbosacral intervertebral disc
N17: Acute kidney failure
Z72: Problems related to lifestyle
Z87: Personal history of other diseases and conditions
Z99: Dependence on enabling machines and devices, not elsewhere c

[2 methods] LightGBM n NTK (n=12)

E03: Other hypothyroidism
E13: Other specified diabetes mellitus
I10: Essential (primary) hypertension
I12: Hypertensive chronic kidney disease
I25: Chronic ischemic heart disease
M25: Other joint disorder, not elsewhere classified
R50: Fever of other and unknown origin
R94: Abnormal results of function studies
Z51: Encounter for other aftercare and medical care
Z82: Family history of certain disabilities and chronic diseases
Z85: Personal history of malignant neoplasm
Z95: Presence of cardiac and vascular implants and grafts

[2 methods] Elastic Net n Recurrence Enrichment (n=7)

F13: Sedative, hypnotic, or anxiolytic related disorders
F15: Other stimulant related disorders
F33: Major depressive disorder, recurrent
F34: Persistent mood [affective] disorders
F39: Unspecified mood [affective] disorder
K31: Other diseases of stomach and duodenum
K50: Crohn's disease [regional enteritis]

[1 methods] Elastic Net (n=36)

A05: Other bacterial foodborne intoxications, not elsewhere class
B17: Other acute viral hepatitis
C54: Malignant neoplasm of corpus uteri
D17: Benign lipomatous neoplasm
E04: Other nontoxic goiter
E10: Type 1 diabetes mellitus
F69: Unspecified disorder of adult personality and behaviour
G30: Alzheimer's disease
G82: Paraplegia (paraparesis) and quadriplegia (quadriparesis)
G91: Hydrocephalus
H66: Suppurative and unspecified otitis media
H90: Conductive and sensorineural hearing loss
J03: Acute tonsillitis
J34: Other and unspecified disorders of nose and nasal sinuses
K42: Umbilical hernia
K81: Cholecystitis
M20: Acquired deformities of fingers and toes
M23: Internal derangement of knee
M65: Synovitis and tenosynovitis
M67: Other disorders of synovium and tendon
M75: Shoulder lesions
M96: Intraoperative and postprocedural complications and disorder
N60: Benign mammary dysplasia
N81: Female genital prolapse
N93: Other abnormal uterine and vaginal bleeding
O02: Other abnormal products of conception
O35: Maternal care for known or suspected fetal abnormality and d
S43: Dislocation and sprain of joints and ligaments of shoulder g
S59: Other and unspecified injuries of elbow and forearm
S82: Fracture of lower leg, including ankle
V85: Occupant of special construction vehicle injured in transpor
W21: Striking against or struck by sports equipment
Z69: Encounter for follow-up examination after completed treatmen
Z39: Encounter for maternal postpartum care and examination
Z45: Encounter for adjustment and management of implanted device
Z47: Orthopedic aftercare

[1 methods] Recurrence Enrichment (n=30)

B37: Candidiasis
B95: Streptococcus, Staphylococcus, and Enterococcus as the cause
B96: Other bacterial agents as the cause of diseases classified e
F20: Schizophrenia
F25: Schizoaffective disorders
F29: Unspecified psychosis not due to a substance or known physio
F60: Specific personality disorders
F90: Attention-deficit hyperactivity disorders
G92: Toxic encephalopathy
J69: Pneumonitis due to solids and liquids
K70: Alcoholic liver disease
K74: Fibrosis and cirrhosis of liver
K76: Other diseases of liver
K85: Acute pancreatitis
L02: Cutaneous abscess, furuncle and carbuncle
L97: Non-pressure chronic ulcer of lower limb, not elsewhere clas
L98: Other disorders of skin and subcutaneous tissue, not elsewhe
M06: Other rheumatoid arthritis
M62: Other disorders of muscle
R40: Somnolence, stupor and coma
R52: Pain, unspecified
R56: Convulsions, not elsewhere classified
R65: Symptoms and signs specifically associated with systemic inf
R69: Unknown and unspecified causes of morbidity
R74: Abnormal serum enzyme levels
R78: Findings of drugs and other substances, not normally found i
T42: Poisoning by, adverse effect of and underdosing of antiepile
T43: Poisoning by, adverse effect of and underdosing of psychotro
T50: Poisoning by, adverse effect of and underdosing of diuretics
Z59: Problems related to housing and economic circumstances

[1 methods] NTK (n=19)

E11: Type 2 diabetes mellitus
I50: Heart failure
I73: Other peripheral vascular diseases
J98: Other respiratory disorders
K29: Gastritis and duodenitis
K92: Other diseases of digestive system
M19: Other and unspecified osteoarthritis
N18: Chronic kidney disease (CKD)
N39: Other disorders of urinary system
R05: Cough
R09: Other symptoms and signs involving the circulatory and respi
R19: Other symptoms and signs involving the digestive system and
R51: Headache
R53: Malaise and fatigue
R55: Syncope and collapse
Z34: Encounter for supervision of normal pregnancy
Z90: Acquired absence of organs, not elsewhere classified
Z96: Presence of other functional implants
Z98: Other postprocedural states

[1 methods] LightGBM (n=11)

D62: Acute posthaemorrhagic anaemia
I44: Atrioventricular and left bundle-branch block
I49: Other cardiac arrhythmias
K63: Other diseases of intestine
N13: Obstructive and reflux uropathy
N40: Hyperplasia of prostate
O80: Encounter for full-term uncomplicated delivery
S72: Fracture of femur
V83: Surgical operation and other surgical procedures as the caus
Y99: External cause status
Z80: Family history of primary malignant neoplasm